Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Yuan, Zeng-Guang Ph.D.		
Project Title:	Effects of Electric Field on Soot Processes in Non-buoyant Hydrocarbon-fueled Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:	Dr. Uday Hegde of National Center for Microgravity Research is the Co-Investigator of this project.		
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	12/01/2000	**End Date:**	11/30/2004
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effects of electric fields on flame shape and size, combustion chemistry, soot formation, and emission are investigated in this project. A microgravity environment is utilized to create a spherically symmetric flame on which a concentric spherical electrical field is superimposed resulting in a one-dimensional configuration. This facilitates study of the interactions between the electric field and the flame. Fuel issues through a spherical burner assembly and burns in the surrounding air to form a spherical diffusion flame. The burner is surrounded by a spherical Faraday cage. The burner and the cage serve as the two electrodes between which the electric field is applied. The overall objective of this project is to establish a fundamental understanding of the effects of imposed electric fields on flame region. These effects include impact on flame sheet location owing to the ionic wind effects and changing in flame chemistry, soot formation and emission, and combustion efficiency. Experimentally obtained data will be used to validate models of the flame processes.		

	Understanding the interaction between the electric field and flame processes is of theoretical, as well as practical, importance. Knowledge about the effects of electric field on flames can be used for flame control, soot reduction and combustion efficiency improvement in a wide variety of combustors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)